

Worksheets W1–W6

accompanying *Mild p -class tower groups of imaginary quadratic fields*

These worksheets contain the data and worked examples referenced from the paper as Parts W1–W6. References of the form Proposition 2.12, §4.2, Section 5, or (P. n) refer to statements, sections, and numbered equations of the paper; plain parenthesised numbers refer to equations of this document.

Part W1: The principal example — class fields and secondary norm operators

This part and Part W2 record the complete data of the first field of the table in Section 5 of the paper. We specialize to

$$K = \mathbf{Q}(\sqrt{-2800905}), \quad p = 5. \quad (1)$$

The field has discriminant

$$\text{disc}(K) = -11203620$$

and

$$\text{Cl}(K) \simeq (\mathbf{Z}/10\mathbf{Z})^3. \quad (2)$$

Fix generators $\kappa_1, \kappa_2, \kappa_3$ for the three cyclic factors in (2). Their images $\overline{\kappa_1}, \overline{\kappa_2}, \overline{\kappa_3}$ in $\text{Cl}(K)/5$ form a basis, and we let

$$a, b, c \in (\text{Cl}(K)/5)^* \simeq H^1(G, \mathbf{F}_5)$$

be the dual character basis, so that for example $a(\overline{\kappa_1}) = 1$ and $a(\overline{\kappa_2}) = a(\overline{\kappa_3}) = 0$. On the torsion side we use

$$e_1 = 2\kappa_1, \quad e_2 = 2\kappa_2, \quad e_3 = 2\kappa_3, \quad (3)$$

which is a basis of $\text{Cl}(K)[5]$. These are separate choices of bases of $\text{Cl}(K)/5$ and $\text{Cl}(K)[5]$; no identification between these two $\mathbf{F}_5$ -vector spaces is being made.

The reconstruction theorem requires the secondary norm operators attached to the three basis characters and their three pairwise sums, namely

$$a, \quad b, \quad c, \quad a+b, \quad a+c, \quad b+c. \quad (4)$$

We first record explicitly the six corresponding unramified cyclic extensions. For a character x , the vector $\boldsymbol{\eta}$ in the second column is the integer character used by PARI on $\text{Cl}(K) \simeq (\mathbf{Z}/10\mathbf{Z})^3$. Since each cyclic factor has order 10, the PARI vector is obtained by multiplying the coordinates of x by $10/5 = 2$. Its kernel is an index-5 subgroup of $\text{Cl}(K)$ whose fixed field is L_x .

The last column lists generators of this subgroup directly, in the fixed coordinates determined by $\kappa_1, \kappa_2, \kappa_3$. Thus, for example, the first row says that the kernel belonging to a is generated by $(5, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ in $(\mathbf{Z}/10\mathbf{Z})^3$.

x	PARI vector $\boldsymbol{\eta}$	generators of the class-field kernel
a	$[2, 0, 0]$	$(5, 0, 0), (0, 1, 0), (0, 0, 1)$
b	$[0, 2, 0]$	$(1, 0, 0), (0, 5, 0), (0, 0, 1)$
c	$[0, 0, 2]$	$(1, 0, 0), (0, 1, 0), (0, 0, 5)$
$a+b$	$[2, 2, 0]$	$(5, 0, 0), (4, 1, 0), (0, 0, 1)$
$a+c$	$[2, 0, 2]$	$(5, 0, 0), (0, 1, 0), (4, 0, 1)$
$b+c$	$[0, 2, 2]$	$(1, 0, 0), (0, 5, 0), (0, 4, 1)$

Each displayed subgroup has index 5, so its class field has degree 5 over K . In all six cases the finite relative discriminant is trivial. Hence the extensions $L_x|K$ are unramified at every finite prime; since K is imaginary quadratic, they are everywhere unramified.

For the Ahlqvist–Carlson formula we need more than the field L_x : the generator $\sigma_x \in \text{Gal}(L_x|K)$ must be normalized by the actual character x . We therefore also record the normalization data used in the computation, in the notation of §4.2 of the paper: σ_{Artin} denotes the cyclic generator of $\text{Gal}(L_x|K)$ chosen by the class-field computation, and the vector $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ is defined by

$$\text{Artin}(\kappa_i) = \sigma_{\text{Artin}}^{\alpha_i}.$$

The scalar λ is characterized by $\alpha_i = \lambda x(\overline{\kappa_i})$ in $\mathbf{F}_5$; the scalar u compares the generator initially used by the implementation with σ_{Artin} , and ν_x is defined by $\sigma_x = \sigma_{\text{Artin}}^{\nu_x}$. Thus ν_x is the exponent of the finally normalized generator. In the present computation $u = 1$ in every case, so $\nu_x = \lambda$.

x	α	λ	u	ν_x
a	$[3, 0, 0]$	3	1	3
b	$[0, 4, 0]$	4	1	4
c	$[0, 0, 4]$	4	1	4
$a + b$	$[3, 3, 0]$	3	1	3
$a + c$	$[3, 0, 3]$	3	1	3
$b + c$	$[0, 2, 2]$	2	1	2

As a direct check of the normalization, for every row and every $i = 1, 2, 3$ we verified

$$\text{Artin}(\kappa_i) = \sigma_x^{x(\overline{\kappa_i})}.$$

Proposition 1 (Secondary norm matrices). *Represent an element of $\text{Cl}(K)/5$ by its coordinate column vector with respect to $\overline{\kappa_1}, \overline{\kappa_2}, \overline{\kappa_3}$. With respect to the input basis e_1, e_2, e_3 of $\text{Cl}(K)[5]$, the six secondary norm operators $D_x: \text{Cl}(K)[5] \rightarrow \text{Cl}(K)/5$ are represented by the matrices*

$$\begin{aligned}
D_a &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 3 & 0 \end{pmatrix}, & D_b &= \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 4 & 2 & 4 \end{pmatrix}, & D_c &= \begin{pmatrix} 1 & 2 & 0 \\ 4 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \\
D_{a+b} &= \begin{pmatrix} 0 & 3 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 2 \end{pmatrix}, & D_{a+c} &= \begin{pmatrix} 1 & 4 & 0 \\ 1 & 4 & 4 \\ 4 & 1 & 0 \end{pmatrix}, & D_{b+c} &= \begin{pmatrix} 3 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 2 & 3 \end{pmatrix}.
\end{aligned}$$

Proof. For each of the 18 pairs (x, e_j) , the constructed pair $(t_{x,j}, I'_{x,j})$ satisfies the two exact equations (P.48)–(P.49). By Proposition 2.12 of the paper, these verified equations imply

$$D_x(e_j) = [N_x(I'_{x,j})] \in \text{Cl}(K)/5.$$

The norm ideal on the right is recomputed independently and its class is expressed in the coordinates $\overline{\kappa_1}, \overline{\kappa_2}, \overline{\kappa_3}$. The resulting values are

x	$D_x(e_1)$	$D_x(e_2)$	$D_x(e_3)$
a	$[0, 0, 0]$	$[0, 3, 3]$	$[0, 1, 0]$
b	$[0, 0, 4]$	$[1, 0, 2]$	$[2, 0, 4]$
c	$[1, 4, 0]$	$[2, 0, 0]$	$[0, 1, 0]$
$a + b$	$[0, 0, 0]$	$[3, 2, 1]$	$[1, 4, 2]$
$a + c$	$[1, 1, 4]$	$[4, 4, 1]$	$[0, 4, 0]$
$b + c$	$[3, 0, 0]$	$[1, 3, 2]$	$[2, 2, 3]$

For example, the second entry in the row for a says

$$D_a(e_2) = 3\overline{\kappa_2} + 3\overline{\kappa_3} \in \text{Cl}(K)/5,$$

and this is the second column of the displayed matrix D_a . Reading the three values in each row as coordinate columns gives the six matrices above. The matrices were assembled only after all norm classes had been computed and independently extracted. $\square$

As an additional arithmetic check, we repeated the complete class-field and secondary-norm computation for the three doubled characters and obtained

$$D_{2a} = 4D_a, \quad D_{2b} = 4D_b, \quad D_{2c} = 4D_c. \quad (5)$$

By the quadratic dependence (P.14) these identities are forced; confirming them establishes nothing new, and their worth lies in the fact that they could have failed. The pairs x and $2x$ define the same class-field kernel but require different normalized automorphisms, and a computation that attached the automorphism to $\ker x$ rather than to x would return $D_{2x} = D_x$ here. The check therefore excludes one specific way for the implementation to go wrong at its most delicate step, and no more. What it probes belongs to the procedure rather than to the field, so we applied this check only to the principal example.

Part W2: The principal example — the transversality computation

We apply Corollary 3.15 of the paper to

$$x = a + b.$$

From the fourth matrix in Proposition 1,

$$\text{rk } D_{a+b} = 1,$$

with

$$\ker D_{a+b} = \langle e_1, 3e_2 + e_3 \rangle$$

and

$$\text{im } D_{a+b} = \langle \overline{\kappa_1} + 4\overline{\kappa_2} + 2\overline{\kappa_3} \rangle.$$

Take $v_1 = a$, $v_2 = c$, and $y = b + 3c$. The character y annihilates the image line above, while its restriction to $(a + b)^\perp$ is nonzero. With the adapted datum

$$\mathcal{B} = (e_1, 3e_2 + e_3; a, c; b + 3c),$$

the transversality matrix is

$$B_{a+b, \mathcal{B}} = \begin{pmatrix} 3 & 2 \\ 3 & 4 \end{pmatrix}, \quad \det B_{a+b, \mathcal{B}} = 1 \in \mathbf{F}_5.$$

Thus $a + b$ is transverse, and Theorem 3.13 of the paper proves that $G_\emptyset(K)(5)$ is mild, and hence has cohomological dimension 2.

As an independent check, the six matrices also reconstruct a cubic word matrix of rank three. A direct change of variables and relation basis gives three relations with combinatorially free leading words

$$\text{ZZX}, \quad \text{ZYY}, \quad \text{ZYG}.$$

The accompanying computation checks the shuffle identities, the word order, and Anick's non-overlap condition directly. These additional calculations are not needed for the preceding proof.

Part W3: The norm-degeneracy scheme of the principal example

The norm-degeneracy scheme of Definition 3.17 of the paper can be written out completely for the principal example. With $x = ua + vb + wc$ in the basis of Part W1, polarizing the six matrices of Proposition 1 gives the matrix of quadratic forms

$$D_{(u,v,w)} = \begin{pmatrix} 2vw + w^2 & 2uv + 2uw + v^2 + 3vw + 2w^2 & 4uv + 2v^2 \\ 2uw + vw + 4w^2 & 3u^2 + 4uv + uw + 3vw & u^2 + 3uv + 2uw + vw + w^2 \\ uv + 4uw + 4v^2 + vw & 3u^2 + uv + 3uw + 2v^2 & 3uv + 4v^2 + 4vw \end{pmatrix},$$

whose j th column is $D_x(e_j)$ in the coordinates $\overline{\kappa_1}, \overline{\kappa_2}, \overline{\kappa_3}$. The scheme $\Sigma_D \subseteq \mathbf{P}_{\mathbf{F}_5}^2$ is defined by the nine 2×2 minors, the homogeneous quartics

$$\begin{aligned} & 2u^2vw + 4u^2w^2 + 4uv^2w + 4uw^3 + 4v^3w + 4v^2w^2 + 4vw^3 + 2w^4, \\ & 4u^2vw + u^2w^2 + 3uv^2w + uvw^2 + 2uw^3 + 3v^3w + 4v^2w^2 + 3vw^3 + w^4, \\ & 2u^3w + 4u^2vw + u^2w^2 + 4uv^2w + uvw^2 + uw^3 + 4v^2w^2 + 2w^4, \\ & 3u^2v^2 + u^2vw + uv^3 + uvw^2 + v^4 + v^3w + v^2w^2 + 3vw^3, \\ & u^2v^2 + 4u^2vw + 2uv^3 + 4uv^2w + 3uvw^2 + 2v^4 + v^3w + 2v^2w^2 + 4vw^3, \\ & 3u^3v + u^2v^2 + 4u^2vw + uv^3 + 4uv^2w + 4uvw^2 + v^3w + 3vw^3, \\ & 2u^3v + 4u^3w + 4u^2v^2 + 4u^2w^2 + 4uv^3 + 4uv^2w + 4uvw^2 + 2uw^3, \\ & 4u^3v + u^3w + 3u^2v^2 + u^2vw + 2u^2w^2 + 3uv^3 + 4uv^2w + 3uvw^2 + uw^3, \\ & 2u^4 + 4u^3v + u^3w + 4u^2v^2 + u^2vw + u^2w^2 + 4uv^2w + 2uw^3. \end{aligned}$$

Its closed points can be listed explicitly. On the chart $w = 1$ the radical of the minor ideal has a reduced coordinate ring of $\mathbf{F}_5$ -dimension three, which a single closed point of degree three exhausts; on the line $w = 0$ the ideal restricts to a squarefree quadratic with two rational roots, and $(1:0:0)$ does not lie on Σ_D . Thus Σ_D has exactly three closed points: the rational points $(1:1:0)$ and $(1:3:0)$ and one point of degree three. At $(1:1:0)$ — the character $a + b$ of Part W2 — the tangent space is zero: the point is reduced and isolated, which is the geometric form (Corollary 3.23 of the paper) of the determinant computation carried out there. At $(1:3:0)$ the operator also has rank one, but the tangent space is one-dimensional, so this point is not transverse: rank one alone does not suffice for mildness. The degree-three point is again reduced and isolated, with transversality over its residue field $\mathbf{F}_{125}$; either of the two transverse points proves Theorem 5.1 of the paper for this field. The display is generated deterministically from the stored matrices by `tools/sigma_d_equations.py`.

Part W4: One certificate entry read in full

To make the certificate format concrete, we follow a single entry from prescribed character to certified value; the companion document `certificate-guide.pdf` generated for this field restates all 27 entries in the same notation. Entry 2 certifies the value $D_a(e_2)$, that is, the second column of the first matrix in Proposition 1.

object	stored data (entry 2)
character $x = a$	value vector $(1, 0, 0)$ on the classes of the three stored generators of $\text{Cl}(K)$
extension L_x	relative polynomial $y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$ over $K = \mathbf{Q}(s)$, $s^2 + 2800905 = 0$; an absolute degree-10 model over $\mathbf{Q}$ is stored alongside. The exact Artin kernel check identifies L_x as the fixed field of $\ker x$
generator σ_x	the coordinate vector of $\sigma_x(\theta)$ in the integral basis of the absolute model, θ its primitive element; the normalization (P.22) is checked through exact Artin exponents
integral bases	the bases those coordinates refer to, one for K and one for L_x . An integral basis is not canonical, so a certificate that omitted them would be readable only where the same ones happened to be produced; the verifier expresses the stored bases in its own and requires the change of basis to be integral with determinant ± 1 , which forces the two to span the same ring of integers
pair (a', J) for e_2	$a' = \frac{41298191}{2498705294406001} + \frac{16828}{2498705294406001} s$ and $J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}$ in Hermite normal form, $N(J) = 1201$; the identity $(a')J^5 = \mathcal{O}_K$ — the condition in (P.4) written multiplicatively — is verified
auxiliary pair (t, I')	I' an ideal of L_x of absolute norm $10587476640 = 2^5 \cdot 3 \cdot 5 \cdot 13 \cdot 1696711$, stored in Hermite normal form; t stored in compact form as a product of 128 factors with signed exponents, and never expanded — its principal ideal and residues are evaluated factor by factor
identity (P.48)	$I'^{(1-\sigma_x)^2}(t) J \mathcal{O}_{L_x} = \mathcal{O}_{L_x}$, verified as an equality of fractional ideals
identity (P.49)	the principal ideal generated by $N_x(t)/a'$ is computed from the compact representation and found to be $\mathcal{O}_K$ — not inferred from (P.48), as in §4.3 of the paper; since the torsion units of K are ± 1 , the reduction at the stored degree-one prime above $\ell = 11$, which equals the residue of $+1$, proves $N_x(t) = a'$
certified value	$[N_x(I')] = (0, 3, 3)$ in the coordinates of $\text{Cl}(K)/5$; by (P.26) this is $D_a(e_2)$, the second column of D_a in Proposition 1

Entries 1–18 certify the six matrices of Proposition 1 column by column in this way; entries 19–27 certify the doubled characters $2a$, $2b$, $2c$ entering the consistency identities.

Part W5: The $p = 5$ table

The 204 imaginary quadratic fields with 5-class rank three and $|D_K| < 2^{30}$. Table 1 lists the 61 fields of the first block $|D_K| < 2^{28}$ in full detail; its columns record the class group, the number of $\mathbf{F}_5$ -rational points of the norm-degeneracy scheme Σ_D (Definition 3.17 of the paper) at which $\text{rk } D_x = 1$, the number of these which are transverse, and the smallest degree of an extension of $\mathbf{F}_5$ over which a transverse point exists. The defining polynomials of the extensions used for the eleven fields without a rational transverse point, the coordinates of all transverse points, and the corresponding transversality matrices are recorded in `examples/p5/transverse-rank-one/report.txt`; the finite-field calculations underlying the table accompany the arithmetic result files.

	D_K	$\text{Cl}(K)$	rank-one over $\mathbf{F}_5$	transverse over $\mathbf{F}_5$	least transverse degree
1	-11203620	[10, 10, 10]	2	1	1
2	-18397407	[40, 10, 5]	3	3	1
3	-27960639	[40, 10, 10]	2	2	1
4	-35663739	[30, 10, 5]	0	0	6
5	-51213139	[75, 5, 5]	1	0	4
6	-54319112	[60, 10, 5]	4	2	1
7	-61040707	[20, 10, 5]	2	1	1
8	-65818135	[30, 10, 10]	3	2	1
9	-75949255	[40, 20, 5]	2	2	1
10	-77778287	[120, 10, 5]	3	3	1
11	-89017304	[170, 10, 5]	2	2	1
12	-89218664	[310, 5, 5]	1	1	1
13	-90903207	[110, 10, 5]	2	2	1
14	-93121640	[90, 10, 10]	1	1	1
15	-104545864	[60, 10, 5]	1	1	1
16	-106660295	[110, 10, 10]	2	2	1
17	-109909943	[100, 10, 10]	1	1	1
18	-123482119	[270, 5, 5]	4	4	1
19	-123779560	[10, 10, 10, 2]	2	1	1
20	-126740891	[195, 5, 5]	0	0	6
21	-129063459	[60, 10, 5]	1	1	1
22	-130370807	[100, 10, 10]	0	0	2
23	-134005207	[100, 10, 5]	3	2	1
24	-145367147	[125, 5, 5]	2	2	1
25	-150535579	[140, 5, 5]	2	2	1
26	-150679519	[150, 10, 5]	0	0	6
27	-150752699	[160, 5, 5]	2	1	1
28	-156145931	[165, 5, 5]	3	3	1
29	-164995035	[60, 10, 5]	1	0	2
30	-165570488	[60, 20, 5]	1	1	1
31	-167094823	[60, 10, 10]	1	1	1
32	-176386616	[180, 10, 5]	2	1	1
33	-181738571	[100, 10, 5]	2	1	1
34	-182240771	[225, 5, 5]	3	3	1
35	-186243704	[40, 20, 10]	0	0	3
36	-187160831	[60, 20, 10]	2	1	1
37	-190572839	[630, 5, 5]	2	2	1
38	-194080808	[70, 10, 10]	1	1	1
39	-199674799	[340, 5, 5]	1	1	1
40	-202331347	[30, 10, 5]	1	1	1
41	-203145767	[60, 10, 10, 2]	2	1	1
42	-203437103	[210, 10, 5]	3	2	1
43	-204909944	[80, 10, 10]	1	1	1
44	-205996583	[100, 10, 10]	1	1	1
45	-207666763	[50, 10, 5]	1	1	1
46	-209149784	[180, 10, 5]	3	2	1
47	-212229543	[60, 10, 10]	1	0	3
48	-216770467	[95, 5, 5]	1	1	1
49	-225826216	[20, 10, 10, 2]	1	1	1
50	-226373428	[30, 10, 10]	3	3	1
51	-231112315	[20, 10, 10]	2	2	1

	D_K	$\text{Cl}(K)$	rank-one over $\mathbf{F}_5$	transverse over $\mathbf{F}_5$	least transverse degree
52	-239452895	[160, 10, 10]	1	1	1
53	-242963687	[370, 5, 5]	0	0	6
54	-246238555	[70, 10, 5]	1	1	1
55	-247127879	[160, 10, 10]	1	1	1
56	-248046051	[60, 10, 5]	1	1	1
57	-249407767	[60, 10, 10]	0	0	6
58	-252735992	[280, 5, 5]	5	4	1
59	-253425979	[135, 5, 5]	0	0	3
60	-256232447	[420, 5, 5]	1	1	1
61	-261286872	[10, 10, 10, 2, 2]	1	1	1

Table 2 lists the 142 mild fields with $2^{28} \leq |D_K| < 2^{30}$, with the class group and the least degree of an extension of $\mathbf{F}_5$ carrying a transverse point. The one remaining field of the range, $D_K = -781922404$ with $\text{Cl}(K) \simeq [70, 10, 10]$, is not listed: its norm-degeneracy scheme consists of a single non-reduced closed point of degree two, so no transverse point exists over any extension.

	D_K	$\text{Cl}(K)$	least transverse degree
1	-272394484	[[150, 10, 5]]	6
2	-280557591	[[150, 10, 10]]	1
3	-282198803	[[120, 10, 5]]	1
4	-285889891	[[60, 10, 5]]	1
5	-294897647	[[100, 10, 10]]	6
6	-300240404	[[250, 10, 5]]	1
7	-308728199	[[820, 5, 5]]	2
8	-311069163	[[60, 10, 5]]	1
9	-312163691	[[130, 10, 5]]	1
10	-313065188	[[140, 10, 5]]	1
11	-330727336	[[150, 5, 5]]	6
12	-334692215	[[80, 10, 10, 2]]	2
13	-342156463	[[200, 10, 5]]	1
14	-343576036	[[220, 5, 5]]	2
15	-344438804	[[300, 10, 5]]	2
16	-345369167	[[160, 10, 10]]	1
17	-348003156	[[70, 10, 10]]	1
18	-351044647	[[280, 5, 5]]	1
19	-360803647	[[310, 5, 5]]	1
20	-364780660	[[70, 10, 10]]	1
21	-368523908	[[20, 20, 10, 2]]	2
22	-416349256	[[40, 10, 10]]	1
23	-417286132	[[110, 10, 5]]	1
24	-429818367	[[120, 10, 10]]	1
25	-433597399	[[240, 10, 5]]	1
26	-434084164	[[300, 5, 5]]	2
27	-439073371	[[70, 10, 5]]	1
28	-443585299	[[100, 10, 5]]	1
29	-444408443	[[120, 10, 5]]	1

	D_K	$\text{Cl}(K)$	least transverse degree
30	-448376516	[[110, 10, 10]]	1
31	-467347512	[[20, 10, 10, 2]]	1
32	-469835647	[[80, 20, 5]]	1
33	-480094015	[[350, 5, 5]]	1
34	-480642151	[[270, 10, 5]]	4
35	-482644063	[[100, 10, 10]]	1
36	-498607924	[[160, 10, 5]]	1
37	-501070231	[[775, 5, 5]]	1
38	-506335479	[[190, 10, 10]]	1
39	-508163803	[[30, 10, 10]]	2
40	-509504379	[[140, 10, 5]]	6
41	-512578628	[[80, 10, 10]]	2
42	-516168299	[[450, 5, 5]]	2
43	-520957412	[[230, 10, 5]]	1
44	-522401819	[[320, 5, 5]]	1
45	-527734248	[[80, 10, 10]]	1
46	-541579031	[280, 10, 10]	1
47	-545873304	[70, 10, 10]	1
48	-558181271	[1095, 5, 5]	1
49	-568232120	[50, 10, 10, 2]	1
50	-573379016	[60, 10, 10, 2]	1
51	-578286527	[110, 10, 10, 2]	1
52	-583175288	[50, 10, 10, 2]	2
53	-587516655	[40, 10, 10, 2, 2]	1
54	-587585540	[60, 10, 10, 2]	1
55	-600487131	[80, 10, 10]	1
56	-611642824	[50, 10, 10]	1
57	-614889363	[20, 20, 10]	1
58	-617556791	[270, 10, 10]	1
59	-618229927	[360, 5, 5]	1
60	-638094767	[730, 5, 5]	1
61	-648249064	[70, 10, 10]	1
62	-648716003	[300, 5, 5]	2
63	-652187816	[130, 10, 10]	1
64	-654932024	[330, 10, 5]	6
65	-659084579	[570, 5, 5]	2
66	-663884564	[140, 10, 10]	1
67	-671772331	[320, 5, 5]	1
68	-672002387	[130, 10, 5]	1
69	-684312527	[600, 10, 5]	4
70	-690429160	[20, 10, 10, 2]	1
71	-694030820	[130, 10, 10]	1
72	-694740327	[240, 10, 5]	6
73	-703985192	[390, 5, 5]	1
74	-716214548	[130, 10, 10]	6
75	-716545112	[310, 10, 5]	1
76	-717538411	[215, 5, 5]	1
77	-719003311	[340, 10, 5]	1
78	-720658367	[390, 10, 5]	1
79	-729506291	[520, 5, 5]	1
80	-730579195	[40, 20, 5]	1
81	-736568551	[705, 5, 5]	1

	D_K	$\text{Cl}(K)$	least transverse degree
82	-747498712	[100, 10, 5]	2
83	-747906091	[210, 5, 5]	1
84	-752704467	[130, 5, 5]	1
85	-753660888	[60, 10, 10]	1
86	-758811723	[110, 10, 5]	1
87	-770289259	[275, 5, 5]	1
88	-792236011	[130, 10, 5]	1
89	-818819903	[320, 10, 5]	1
90	-822797303	[480, 10, 5]	1
91	-823763252	[200, 10, 5]	6
92	-829075271	[1490, 5, 5]	1
93	-832956936	[50, 10, 10, 2]	1
94	-834867768	[30, 10, 10, 2]	1
95	-835199035	[50, 10, 10]	1
96	-837148756	[150, 10, 5]	3
97	-839189588	[160, 10, 10]	1
98	-847946303	[1020, 5, 5]	3
99	-848798407	[375, 5, 5]	1
100	-856365643	[170, 5, 5]	1
101	-859104103	[230, 10, 5]	3
102	-860862356	[710, 5, 5]	1
103	-860938607	[390, 10, 5]	2
104	-863732452	[90, 10, 5]	1
105	-864855431	[60, 60, 10]	1
106	-865276015	[280, 10, 5]	3
107	-865288212	[60, 10, 10, 2]	1
108	-865931992	[60, 10, 10]	1
109	-873927415	[240, 10, 5]	1
110	-876422059	[265, 5, 5]	4
111	-880112408	[270, 10, 5]	1
112	-880342436	[660, 5, 5]	1
113	-881852344	[80, 20, 5]	1
114	-882321235	[130, 10, 5]	6
115	-884383779	[180, 10, 5]	1
116	-890032871	[750, 10, 5]	1
117	-912814943	[600, 10, 5]	1
118	-930871220	[160, 10, 10]	6
119	-937521204	[60, 20, 10]	1
120	-941867380	[180, 10, 5]	1
121	-954227011	[270, 5, 5]	6
122	-964632455	[1180, 5, 5]	1
123	-967077539	[450, 5, 5]	1
124	-975828939	[270, 5, 5]	1
125	-976016443	[210, 5, 5]	1
126	-979106379	[410, 5, 5]	6
127	-996554680	[60, 10, 10]	2
128	-1007496479	[920, 10, 5]	1
129	-1008688623	[850, 5, 5]	4
130	-1011574052	[60, 20, 10]	3
131	-1014273539	[360, 5, 5]	6
132	-1016848324	[300, 5, 5]	1
133	-1018718571	[180, 10, 5]	1

	D_K	$\text{Cl}(K)$	least transverse degree
134	-1019215031	[450, 15, 5]	6
135	-1030705032	[40, 10, 10, 2]	1
136	-1040930248	[120, 10, 5]	1
137	-1041244847	[470, 10, 5]	2
138	-1043658971	[550, 5, 5]	2
139	-1047634431	[1030, 5, 5]	1
140	-1064940171	[180, 10, 5]	1
141	-1066461896	[330, 10, 5]	6
142	-1069759951	[360, 10, 5]	1

Part W6: The $p = 3$ table

The 378 mild fields at $p = 3$ with $|D_K| < 2^{29}$, with the class group, the rank of δ , and the source of mildness: Table 3 lists the 151 mild fields of the first block $|D_K| < 2^{28}$, Table 4 the 227 mild fields with $2^{28} \leq |D_K| < 2^{29}$. Additional records: the Anick witnesses in `examples/p3/results/block-witnesses-002.json`; the Gröbner verdicts, including the seven fields that are provably not strongly free, in `examples/p3/results/strong-freeness-block-001.json`; the cone-criterion report with the transverse cone point of every criterion field in `examples/p3/cone-criterion/report.txt`, with the per-field summary `examples/p3/cone-criterion/summary.json`. The drivers are `examples/p3/strong-freeness/block_sweep.py`, `tools/strong_freeness_singular.py`, and `tools/p3_cone_criterion.py`. The exhaustive Anick searches over the two subfamilies of Section 6 of the paper ran over all 11 232 changes of variables over $\mathbf{F}_3$ and all 663 390 ordered projective bases over $\mathbf{F}_9$.

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
1	-4447704	[24, 6, 6]	3	Hilbert series
2	-4897363	[33, 3, 3]	3	Hilbert series
3	-11967780	[18, 6, 6, 2]	2	transverse point, degree 1
4	-16564388	[36, 12, 3]	2	transverse point, degree 1
5	-19006932	[18, 6, 6, 2]	2	transverse point, degree 1
6	-19941763	[72, 3, 3]	2	transverse point, degree 1
7	-25680535	[72, 6, 6]	2	transverse point, degree 1
8	-29225219	[216, 3, 3]	2	Hilbert series
9	-33886715	[72, 6, 3]	2	transverse point, degree 1
10	-37882679	[396, 6, 3]	2	transverse point, degree 1
11	-38295992	[108, 6, 3]	2	transverse point, degree 1
12	-42305279	[414, 6, 3]	2	Hilbert series
13	-45152879	[816, 3, 3]	3	Hilbert series
14	-48023907	[72, 6, 3]	2	transverse point, degree 1
15	-50179656	[84, 6, 6]	3	Hilbert series
16	-52526399	[990, 3, 3]	2	transverse point, degree 1
17	-53209523	[81, 9, 3]	1	transverse point, degree 1
18	-67271572	[72, 6, 3]	2	Hilbert series
19	-68523256	[72, 12, 3]	2	transverse point, degree 1
20	-69456767	[819, 3, 3]	2	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
21	-70164019	[189, 3, 3]	2	transverse point, degree 1
22	-70488707	[204, 3, 3]	3	Hilbert series
23	-78616756	[72, 6, 6]	2	transverse point, degree 1
24	-79446183	[618, 3, 3]	3	Hilbert series
25	-80667656	[408, 6, 3]	3	Hilbert series
26	-81592963	[117, 3, 3]	2	transverse point, degree 1
27	-87155515	[24, 12, 6]	3	Hilbert series
28	-87893823	[414, 6, 3]	2	Hilbert series
29	-92956243	[144, 3, 3]	2	transverse point, degree 1
30	-98512820	[144, 6, 6]	2	Hilbert series
31	-101375499	[72, 9, 3]	1	transverse point, degree 1
32	-105273787	[144, 3, 3]	2	transverse point, degree 1
33	-105274283	[297, 3, 3]	2	Hilbert series
34	-105569787	[72, 6, 3]	2	transverse point, degree 1
35	-106331556	[138, 6, 3]	3	Hilbert series
36	-107033560	[144, 6, 3]	2	transverse point, degree 1
37	-110528935	[150, 6, 6]	3	Hilbert series
38	-112456564	[126, 6, 6]	2	transverse point, degree 1
39	-112596647	[516, 3, 3]	3	Hilbert series
40	-113167511	[1239, 3, 3]	3	Hilbert series
41	-115676840	[96, 6, 6, 2]	3	Hilbert series
42	-119808287	[846, 3, 3]	2	transverse point, degree 1
43	-120340607	[1026, 3, 3]	2	transverse point, degree 1
44	-123939723	[252, 3, 3]	2	transverse point, degree 1
45	-124171604	[72, 12, 6]	2	transverse point, degree 1
46	-128631659	[444, 3, 3]	3	Hilbert series
47	-129794180	[432, 6, 3]	2	transverse point, degree 1
48	-131071559	[1575, 3, 3]	2	Hilbert series
49	-132696427	[225, 3, 3]	2	transverse point, degree 1
50	-132768840	[126, 6, 6]	2	transverse point, degree 1
51	-133858164	[60, 6, 6, 2]	3	Hilbert series
52	-134034647	[90, 18, 6]	1	transverse point, degree 1
53	-136185871	[1080, 3, 3]	2	Hilbert series
54	-137034408	[90, 6, 6]	2	transverse point, degree 1
55	-137225371	[396, 3, 3]	2	transverse point, degree 1
56	-137752583	[1173, 3, 3]	3	Hilbert series
57	-138230347	[90, 9, 3]	1	transverse point, degree 1
58	-139272611	[207, 9, 3]	1	transverse point, degree 2
59	-139854279	[252, 12, 3]	2	transverse point, degree 1
60	-140436319	[810, 3, 3]	2	Hilbert series
61	-141590039	[1854, 3, 3]	2	transverse point, degree 1
62	-142082007	[270, 6, 3]	2	Hilbert series
63	-143978071	[837, 3, 3]	2	transverse point, degree 1
64	-147364807	[387, 3, 3]	2	transverse point, degree 1
65	-147372692	[186, 6, 6]	3	Hilbert series
66	-147994487	[432, 9, 3]	1	transverse point, degree 1
67	-149548296	[144, 6, 6]	2	transverse point, degree 1
68	-157432807	[558, 3, 3]	2	transverse point, degree 1
69	-159218551	[798, 3, 3]	3	Hilbert series
70	-159842587	[165, 3, 3]	3	Hilbert series
71	-159949967	[756, 3, 3]	2	transverse point, degree 1
72	-163004039	[252, 18, 3]	1	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
73	-164195095	[162, 6, 6]	2	transverse point, degree 1
74	-164443092	[84, 6, 6]	3	Hilbert series
75	-164490995	[96, 6, 6]	3	Hilbert series
76	-165028287	[252, 6, 3]	2	transverse point, degree 1
77	-165939880	[36, 6, 6, 2]	2	Hilbert series
78	-166596251	[279, 9, 3]	1	transverse point, degree 1
79	-168380088	[42, 6, 6, 2]	3	Hilbert series
80	-168657416	[324, 6, 3]	2	transverse point, degree 1
81	-173453135	[540, 6, 3]	2	transverse point, degree 1
82	-177327572	[426, 3, 3]	3	Hilbert series
83	-177661288	[90, 6, 6]	2	transverse point, degree 1
84	-177907703	[1098, 3, 3]	2	transverse point, degree 1
85	-177911939	[483, 3, 3]	3	Hilbert series
86	-179262491	[228, 6, 3]	3	Hilbert series
87	-181950799	[738, 3, 3]	2	transverse point, degree 1
88	-184537012	[138, 6, 3]	3	Hilbert series
89	-186324791	[924, 6, 3]	3	Hilbert series
90	-187191391	[843, 3, 3]	3	Hilbert series
91	-188352419	[840, 3, 3]	3	Hilbert series
92	-188424655	[180, 18, 3]	1	Hilbert series
93	-192103243	[126, 6, 3]	2	transverse point, degree 1
94	-192928619	[126, 6, 6]	2	Hilbert series
95	-193481043	[126, 6, 3]	2	transverse point, degree 1
96	-193816927	[795, 3, 3]	3	Hilbert series
97	-194921531	[630, 3, 3]	2	transverse point, degree 1
98	-198040904	[396, 9, 3]	1	transverse point, degree 1
99	-198494239	[891, 3, 3]	2	transverse point, degree 1
100	-198524584	[486, 3, 3]	2	transverse point, degree 1
101	-204084679	[828, 3, 3]	2	transverse point, degree 1
102	-206568676	[162, 6, 6]	2	Hilbert series
103	-207079079	[1422, 3, 3]	2	transverse point, degree 1
104	-207482084	[720, 6, 3]	2	Hilbert series
105	-210370495	[90, 6, 6, 2]	2	transverse point, degree 1
106	-211248887	[1188, 3, 3]	2	Anick witness
107	-211982516	[414, 6, 3]	2	transverse point, degree 1
108	-213255839	[648, 6, 6]	2	Hilbert series
109	-213954164	[486, 6, 3]	2	transverse point, degree 1
110	-214829931	[396, 3, 3]	2	transverse point, degree 1
111	-215588015	[1926, 3, 3]	2	transverse point, degree 1
112	-220005619	[228, 3, 3]	3	Hilbert series
113	-221338583	[1134, 3, 3]	2	Hilbert series
114	-224238388	[234, 6, 3]	2	transverse point, degree 1
115	-225512103	[414, 6, 3]	2	transverse point, degree 1
116	-226566408	[360, 6, 3]	2	transverse point, degree 1
117	-228404408	[180, 9, 3]	1	transverse point, degree 1
118	-229531859	[636, 3, 3]	3	Hilbert series
119	-229988136	[84, 6, 6, 2]	3	Hilbert series
120	-230059147	[258, 3, 3]	3	Hilbert series
121	-230514583	[342, 6, 3]	2	transverse point, degree 1
122	-231693143	[1512, 3, 3]	2	transverse point, degree 1
123	-232873860	[72, 6, 6, 2]	2	transverse point, degree 1
124	-234571683	[72, 12, 3]	2	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
125	-235997807	[1467, 3, 3]	2	transverse point, degree 1
126	-236414323	[384, 3, 3]	3	Hilbert series
127	-237129972	[216, 6, 3]	2	transverse point, degree 1
128	-239565368	[192, 12, 3]	3	Hilbert series
129	-242115347	[498, 3, 3]	3	Hilbert series
130	-242584547	[564, 3, 3]	3	Hilbert series
131	-242722852	[144, 6, 3]	2	transverse point, degree 1
132	-243949704	[360, 6, 3]	2	Hilbert series
133	-245576008	[396, 3, 3]	2	transverse point, degree 1
134	-245591416	[126, 6, 6]	2	transverse point, degree 1
135	-246038399	[2952, 3, 3]	2	transverse point, degree 1
136	-247085519	[456, 6, 6]	3	Hilbert series
137	-249699611	[666, 3, 3]	2	transverse point, degree 1
138	-249851636	[630, 6, 3]	2	transverse point, degree 1
139	-250419887	[1053, 3, 3]	2	transverse point, degree 1
140	-251244319	[246, 6, 6]	3	Hilbert series
141	-251525471	[1953, 3, 3]	2	transverse point, degree 1
142	-255182655	[126, 6, 6, 2]	2	transverse point, degree 1
143	-256421332	[414, 3, 3]	2	Hilbert series
144	-256600003	[198, 3, 3]	2	transverse point, degree 1
145	-257735579	[972, 3, 3]	2	transverse point, degree 1
146	-263310215	[174, 6, 6, 2]	3	Hilbert series
147	-263329352	[90, 6, 6, 2]	2	transverse point, degree 1
148	-263780072	[234, 6, 6]	2	Anick witness
149	-264862580	[456, 6, 3]	3	Hilbert series
150	-264874404	[282, 6, 3]	3	Hilbert series
151	-265389256	[288, 6, 3]	2	transverse point, degree 1

Table 4: the 227 mild fields of the second block. The twelve fields of the range that are provably not strongly free are $D_K = -54962868, -123927735, -135819467, -176214724, -252545119, -257444104, -259111511$ (first block) and $D_K = -297353236, -306860520, -400506295, -522302531, -526671699$ (second block).

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
1	-269035079	[2742, 3, 3]	3	Hilbert series
2	-269442491	[936, 3, 3]	2	transverse point, degree 1
3	-270840455	[2100, 3, 3]	3	Hilbert series
4	-271882751	[558, 6, 6]	2	transverse point, degree 1
5	-272531631	[180, 18, 3]	1	transverse point, degree 1
6	-273772679	[2412, 3, 3]	2	transverse point, degree 1
7	-276000263	[774, 6, 3]	2	transverse point, degree 1
8	-276510583	[720, 3, 3]	2	transverse point, degree 1
9	-278078212	[72, 18, 3]	1	transverse point, degree 1
10	-278248660	[114, 6, 6]	3	Hilbert series
11	-278854280	[162, 18, 3]	1	transverse point, degree 3
12	-281764488	[252, 6, 3]	2	transverse point, degree 1
13	-284311556	[252, 6, 6]	2	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
14	-284471583	[84, 6, 6, 2]	3	Hilbert series
15	-287069012	[1026, 3, 3]	2	transverse point, degree 1
16	-287520315	[72, 18, 3]	1	transverse point, degree 1
17	-288780756	[102, 6, 6, 2]	3	Hilbert series
18	-289430627	[282, 6, 3]	3	Hilbert series
19	-289767144	[180, 6, 6]	2	transverse point, degree 1
20	-289949683	[207, 3, 3]	2	transverse point, degree 1
21	-290892479	[2016, 3, 3]	2	transverse point, degree 1
22	-292462495	[162, 18, 3]	1	transverse point, degree 2
23	-294047576	[198, 18, 3]	1	transverse point, degree 1
24	-296567203	[108, 6, 3]	2	transverse point, degree 1
25	-297292999	[1719, 3, 3]	2	Hilbert series
26	-300555103	[756, 3, 3]	2	transverse point, degree 1
27	-301447480	[234, 6, 3]	2	Hilbert series
28	-303172043	[108, 12, 3]	2	transverse point, degree 1
29	-304019767	[675, 3, 3]	2	transverse point, degree 1
30	-304716855	[276, 6, 6]	3	Hilbert series
31	-306171172	[396, 3, 3]	2	transverse point, degree 1
32	-309167771	[270, 6, 6]	2	Hilbert series
33	-309523651	[378, 3, 3]	2	transverse point, degree 1
34	-312889363	[276, 3, 3]	3	Hilbert series
35	-315192871	[582, 6, 3]	3	Hilbert series
36	-315388527	[642, 6, 3]	3	Hilbert series
37	-315578903	[783, 9, 3]	1	transverse point, degree 1
38	-316179640	[162, 6, 6]	2	Hilbert series
39	-316236695	[648, 9, 3]	1	transverse point, degree 1
40	-317103719	[180, 36, 3]	1	transverse point, degree 1
41	-319277695	[234, 6, 6]	2	Hilbert series
42	-320017411	[114, 6, 6]	3	Hilbert series
43	-320230023	[846, 3, 3]	2	transverse point, degree 1
44	-320873055	[414, 6, 6]	2	transverse point, degree 1
45	-321821807	[1863, 3, 3]	2	transverse point, degree 1
46	-321969236	[774, 6, 3]	2	transverse point, degree 1
47	-322368555	[54, 18, 3]	1	transverse point, degree 1
48	-326265215	[2394, 3, 3]	2	transverse point, degree 1
49	-327730987	[252, 3, 3]	2	Hilbert series
50	-328127039	[2082, 3, 3]	3	Hilbert series
51	-328260664	[468, 3, 3]	2	transverse point, degree 1
52	-328455255	[450, 6, 3]	2	Hilbert series
53	-329114036	[306, 6, 6]	2	transverse point, degree 1
54	-329492855	[1422, 6, 3]	2	transverse point, degree 1
55	-330550647	[216, 6, 6]	2	Hilbert series
56	-333047959	[1710, 3, 3]	2	transverse point, degree 1
57	-333351412	[378, 3, 3]	2	transverse point, degree 1
58	-334299103	[1149, 3, 3]	3	Hilbert series
59	-335378399	[1392, 6, 3]	3	Hilbert series
60	-337706651	[810, 3, 3]	2	transverse point, degree 1
61	-338981272	[72, 6, 6]	2	Hilbert series
62	-340431839	[1032, 6, 3]	3	Hilbert series
63	-340736027	[144, 9, 3]	1	transverse point, degree 1
64	-342388799	[1899, 3, 3]	2	transverse point, degree 1
65	-343225819	[522, 3, 3]	2	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
66	-343795435	[54, 18, 3]	1	Hilbert series
67	-345822859	[531, 3, 3]	2	Hilbert series
68	-345910216	[72, 18, 3]	1	transverse point, degree 1
69	-346173311	[1296, 6, 3]	2	Hilbert series
70	-349147803	[168, 6, 3]	3	Hilbert series
71	-351541032	[162, 6, 6]	2	Hilbert series
72	-352228711	[558, 6, 3]	2	Hilbert series
73	-353213668	[90, 6, 6]	2	transverse point, degree 1
74	-353453432	[792, 3, 3]	2	transverse point, degree 1
75	-354407531	[198, 6, 6]	2	transverse point, degree 1
76	-355084468	[72, 6, 6]	2	transverse point, degree 1
77	-355823027	[702, 3, 3]	2	Hilbert series
78	-357386612	[162, 6, 6]	2	transverse point, degree 1
79	-357593707	[261, 3, 3]	2	transverse point, degree 1
80	-359247935	[1854, 3, 3]	2	transverse point, degree 1
81	-361883812	[306, 6, 3]	2	transverse point, degree 1
82	-362801240	[270, 6, 6]	2	Hilbert series
83	-362841123	[180, 6, 3]	2	transverse point, degree 1
84	-363298004	[1242, 3, 3]	2	Hilbert series
85	-363463435	[342, 3, 3]	2	Hilbert series
86	-364435991	[387, 9, 9]	0	transverse point, degree 1
87	-368906059	[315, 3, 3]	2	transverse point, degree 1
88	-369006184	[408, 6, 3]	3	Hilbert series
89	-373392228	[90, 6, 6, 2]	2	transverse point, degree 1
90	-373612983	[630, 6, 3]	2	transverse point, degree 1
91	-376723499	[990, 3, 3]	2	Hilbert series
92	-377080952	[378, 6, 6]	2	transverse point, degree 1
93	-379723327	[675, 3, 3]	2	Hilbert series
94	-380426767	[516, 6, 3]	3	Hilbert series
95	-384651719	[1848, 3, 3]	3	Hilbert series
96	-385739807	[2532, 3, 3]	3	Hilbert series
97	-387053719	[1692, 3, 3]	2	Hilbert series
98	-388794099	[180, 6, 3]	2	Hilbert series
99	-391022655	[144, 6, 6, 2]	2	Hilbert series
100	-391370664	[288, 6, 6]	2	transverse point, degree 1
101	-392235755	[318, 6, 3]	3	Hilbert series
102	-392418863	[396, 9, 3]	1	transverse point, degree 1
103	-392551903	[348, 6, 3]	3	Hilbert series
104	-393305243	[576, 3, 3]	2	Hilbert series
105	-393832203	[330, 3, 3]	3	Hilbert series
106	-396117535	[36, 18, 6, 2]	1	transverse point, degree 1
107	-397450612	[108, 6, 6]	2	Hilbert series
108	-397620967	[684, 3, 3]	2	Hilbert series
109	-398472692	[420, 6, 3]	3	Hilbert series
110	-402426383	[1332, 3, 3]	2	transverse point, degree 1
111	-405220571	[420, 6, 3]	3	Hilbert series
112	-405289839	[468, 12, 3]	2	transverse point, degree 1
113	-405618691	[396, 3, 3]	2	transverse point, degree 1
114	-405868619	[702, 3, 3]	2	Hilbert series
115	-406709967	[612, 6, 3]	2	transverse point, degree 1
116	-406844851	[222, 6, 3]	3	Hilbert series
117	-407054247	[702, 6, 3]	2	Hilbert series

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
118	-407737032	[192, 6, 6]	3	Hilbert series
119	-409562659	[126, 6, 6]	2	transverse point, degree 1
120	-411135668	[240, 6, 6]	3	Hilbert series
121	-411492291	[120, 6, 6]	3	Hilbert series
122	-411605576	[324, 9, 3]	1	transverse point, degree 1
123	-412074548	[360, 6, 3]	2	transverse point, degree 1
124	-412229751	[1242, 6, 3]	2	Hilbert series
125	-412411215	[468, 6, 6]	2	Hilbert series
126	-412626459	[252, 6, 3]	2	transverse point, degree 1
127	-413263288	[54, 18, 3]	1	transverse point, degree 1
128	-413771887	[243, 9, 3]	1	transverse point, degree 1
129	-414248323	[240, 3, 3]	3	Hilbert series
130	-415404760	[126, 6, 6]	2	transverse point, degree 1
131	-415460063	[1674, 3, 3]	2	transverse point, degree 1
132	-415865399	[3474, 3, 3]	2	transverse point, degree 1
133	-418003771	[72, 18, 3]	1	transverse point, degree 1
134	-418220231	[726, 6, 6]	3	Hilbert series
135	-418591263	[918, 6, 3]	2	transverse point, degree 1
136	-419498659	[252, 6, 3]	2	Hilbert series
137	-423547912	[456, 3, 3]	3	Hilbert series
138	-426349551	[378, 6, 6]	2	transverse point, degree 1
139	-426397188	[24, 12, 6, 2, 2]	3	Hilbert series
140	-429673999	[1062, 6, 3]	2	transverse point, degree 1
141	-429704363	[288, 6, 3]	2	transverse point, degree 1
142	-429817539	[144, 12, 3]	2	Hilbert series
143	-432285639	[198, 18, 3]	1	transverse point, degree 1
144	-433041755	[192, 12, 3]	3	Hilbert series
145	-433265239	[1332, 3, 3]	2	transverse point, degree 1
146	-434995336	[288, 12, 3]	2	transverse point, degree 1
147	-435537911	[3204, 3, 3]	2	transverse point, degree 1
148	-437458168	[48, 12, 6]	3	Hilbert series
149	-438001735	[660, 6, 3]	3	Hilbert series
150	-439077883	[435, 3, 3]	3	Hilbert series
151	-439993003	[252, 3, 3]	2	transverse point, degree 1
152	-441621695	[882, 6, 3]	2	transverse point, degree 1
153	-442137480	[72, 6, 6, 2]	2	transverse point, degree 1
154	-443078148	[390, 6, 3]	3	Hilbert series
155	-443567667	[396, 3, 3]	2	transverse point, degree 1
156	-444251239	[1485, 3, 3]	2	transverse point, degree 1
157	-445630168	[342, 3, 3]	2	transverse point, degree 1
158	-445727263	[684, 3, 3]	2	transverse point, degree 1
159	-448664599	[1440, 3, 3]	2	transverse point, degree 1
160	-450225779	[396, 9, 3]	1	transverse point, degree 1
161	-452043935	[468, 6, 6]	2	transverse point, degree 1
162	-452315803	[54, 18, 3]	1	transverse point, degree 1
163	-452708443	[132, 6, 3]	3	Hilbert series
164	-453787663	[1314, 3, 3]	2	transverse point, degree 1
165	-453940115	[234, 6, 6]	2	transverse point, degree 1
166	-455141263	[456, 6, 3]	3	Hilbert series
167	-455500919	[2727, 3, 3]	2	transverse point, degree 1
168	-456031591	[990, 3, 3]	2	Hilbert series
169	-456392723	[819, 3, 3]	2	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
170	-456750443	[594, 3, 3]	2	transverse point, degree 1
171	-457209563	[666, 3, 3]	2	transverse point, degree 1
172	-458005971	[144, 12, 3]	2	Hilbert series
173	-461217256	[186, 6, 6]	3	Hilbert series
174	-462746308	[576, 3, 3]	2	transverse point, degree 1
175	-464748707	[531, 3, 3]	2	Hilbert series
176	-465578067	[132, 12, 3]	3	Hilbert series
177	-471807499	[369, 3, 3]	2	transverse point, degree 1
178	-473656199	[2505, 3, 3]	3	Hilbert series
179	-474132487	[1062, 3, 3]	2	Hilbert series
180	-474619503	[546, 6, 3]	3	Hilbert series
181	-475603723	[144, 6, 3]	2	transverse point, degree 1
182	-476261771	[1086, 3, 3]	3	Hilbert series
183	-488080687	[432, 9, 3]	1	transverse point, degree 1
184	-491257927	[477, 9, 3]	1	transverse point, degree 1
185	-491588315	[150, 6, 6]	3	Hilbert series
186	-492588391	[324, 12, 3]	2	transverse point, degree 1
187	-493065527	[1815, 3, 3]	3	Hilbert series
188	-493089688	[114, 6, 6]	3	Hilbert series
189	-493620899	[1056, 3, 3]	3	Hilbert series
190	-496813127	[2049, 3, 3]	3	Hilbert series
191	-499897432	[228, 6, 3]	3	Hilbert series
192	-500454835	[216, 6, 3]	2	transverse point, degree 1
193	-501362803	[318, 3, 3]	3	Hilbert series
194	-502048363	[270, 3, 3]	2	transverse point, degree 1
195	-502995268	[288, 9, 3]	1	transverse point, degree 1
196	-503888863	[864, 3, 3]	2	transverse point, degree 1
197	-508498783	[432, 6, 3]	2	transverse point, degree 1
198	-508572472	[198, 6, 3]	2	Hilbert series
199	-508661864	[810, 6, 3]	2	Hilbert series
200	-508992367	[426, 6, 3]	3	Hilbert series
201	-509395507	[360, 3, 3]	2	transverse point, degree 1
202	-509943391	[1575, 3, 3]	2	transverse point, degree 1
203	-509992863	[1530, 3, 3]	2	transverse point, degree 1
204	-510075451	[360, 3, 3]	2	transverse point, degree 1
205	-511192643	[390, 6, 3]	3	Hilbert series
206	-512094407	[1878, 3, 3]	3	Hilbert series
207	-512450420	[360, 6, 6]	2	transverse point, degree 1
208	-513440036	[1464, 3, 3]	3	Hilbert series
209	-515085123	[48, 12, 6]	3	Hilbert series
210	-517651224	[378, 6, 6]	2	Hilbert series
211	-519460644	[342, 6, 6]	2	transverse point, degree 1
212	-519470164	[396, 6, 3]	2	Hilbert series
213	-519793739	[972, 3, 3]	2	transverse point, degree 1
214	-520681447	[288, 6, 6]	2	transverse point, degree 1
215	-522900139	[501, 3, 3]	3	Hilbert series
216	-523483639	[252, 6, 6, 2]	2	transverse point, degree 1
217	-523819956	[108, 6, 6, 2]	2	Hilbert series
218	-525023563	[324, 3, 3]	2	transverse point, degree 1
219	-525305391	[306, 6, 6, 2]	2	transverse point, degree 1
220	-526209032	[288, 12, 3]	2	transverse point, degree 1
221	-526255239	[2520, 3, 3]	2	transverse point, degree 1

	D_K	$\text{Cl}(K)$	$\text{rk } \delta$	mild via
222	-526633655	[900, 6, 3]	2	transverse point, degree 1
223	-527192591	[540, 18, 3]	1	transverse point, degree 1
224	-528544712	[492, 6, 3]	3	Hilbert series
225	-530384824	[900, 3, 3]	2	transverse point, degree 1
226	-534599480	[666, 6, 3]	2	transverse point, degree 1
227	-536670899	[954, 3, 3]	2	transverse point, degree 1